

A COLLABORATION BETWEEN
ANDERSON SERANGOON JUNIOR COLLEGE &
NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 3

CANDIDATE
NAME

CENTRE
NUMBER

CHEMISTRY

9813/01

Paper 1

22 September 2020

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Data Booklet
 Insert

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer all questions in the spaces provided on the Question Paper. If additional space is required, you should use the pages at the end of this booklet. The question number must be clearly shown.

Section A

Answer **all** questions.

Section B

Answer **two** questions.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Setters

Q1: NYJC

Q2: ASRJC

Q3: ASRJC

Q4: ASRJC

Q5: ASRJC

Q6: NYJC

Q7: NYJC

This document consists of **44** printed pages and **1 insert**

Section A

Q1 Setter: NYJC

1 (a) CO_2 has low reactivity and high stability (thermodynamically stable and kinetically inert) which generally limits the reaction with other compounds.

(b) (i) Amount of $\text{CO}_2 = \frac{(0.34 \times 10^6)(40 \times 10^{-6})}{8.31 \times (30+273)} = 5.401 \times 10^{-3} \text{ mol}$

Tertiary amine is the limiting reagent.

Amount of tertiary amine = $\frac{5.401 \times 10^{-3}}{3.6} = 1.500 \times 10^{-3} \text{ mol}$

Mass of *p*-terphenyl = $1.500 \times 10^{-3} \times 20\% \times (18 \times 12.0 + 14 \times 1.0) = 0.0690 \text{ g}$

- (ii) • *p*-terphenyl is a catalyst.
• It is used up in step 1 and regenerated in step 3
• It is used in small amount (20 mol%)

(c) (i) $E(p\text{-terphenyl} / p\text{-terphenyl}^{\bullet-}) = (-2.63) + (+0.244) = -2.386 \text{ V}$
 $E(\text{CO}_2 / \text{CO}_2^{\bullet-}) = (-2.21) + (+0.244) = -1.966 \text{ V}$

(ii) $E_{\text{cell}} = (-1.966) - (-2.386) = +0.42 \text{ V} (> 0, \text{spontaneous})$
or $E_{\text{cell}} = (-2.21) - (-2.63) = +0.42 \text{ V} (> 0, \text{spontaneous})$

$\Delta G = -(1)(96500)(0.42) = -40530 \text{ J mol}^{-1} = -40.5 \text{ kJ mol}^{-1}$



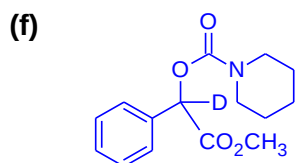
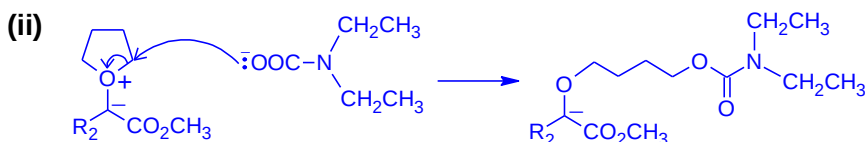
Bond angle = 138°

(d) (i) The *singlet state* experience repulsion between two electrons in the same sp^2 hybrid orbital. In the *triplet state*, the two unpaired electrons are in two different orbitals (p and sp^2 orbital), which experiences lesser repulsion.

(ii) The presence of an empty p orbital in the *singlet state* allows it to accept a lone pair of electrons from the carbamate anion to form the intermediate anion but in the *triplet state*, there is already an unpaired electron in both the p and sp^2 hybrid orbital and unable to accept a lone pair of electrons.

(e) (i) Nucleophilic addition.

CO_2 is an electrophile as it accepts a lone pair of electrons from the secondary amine.



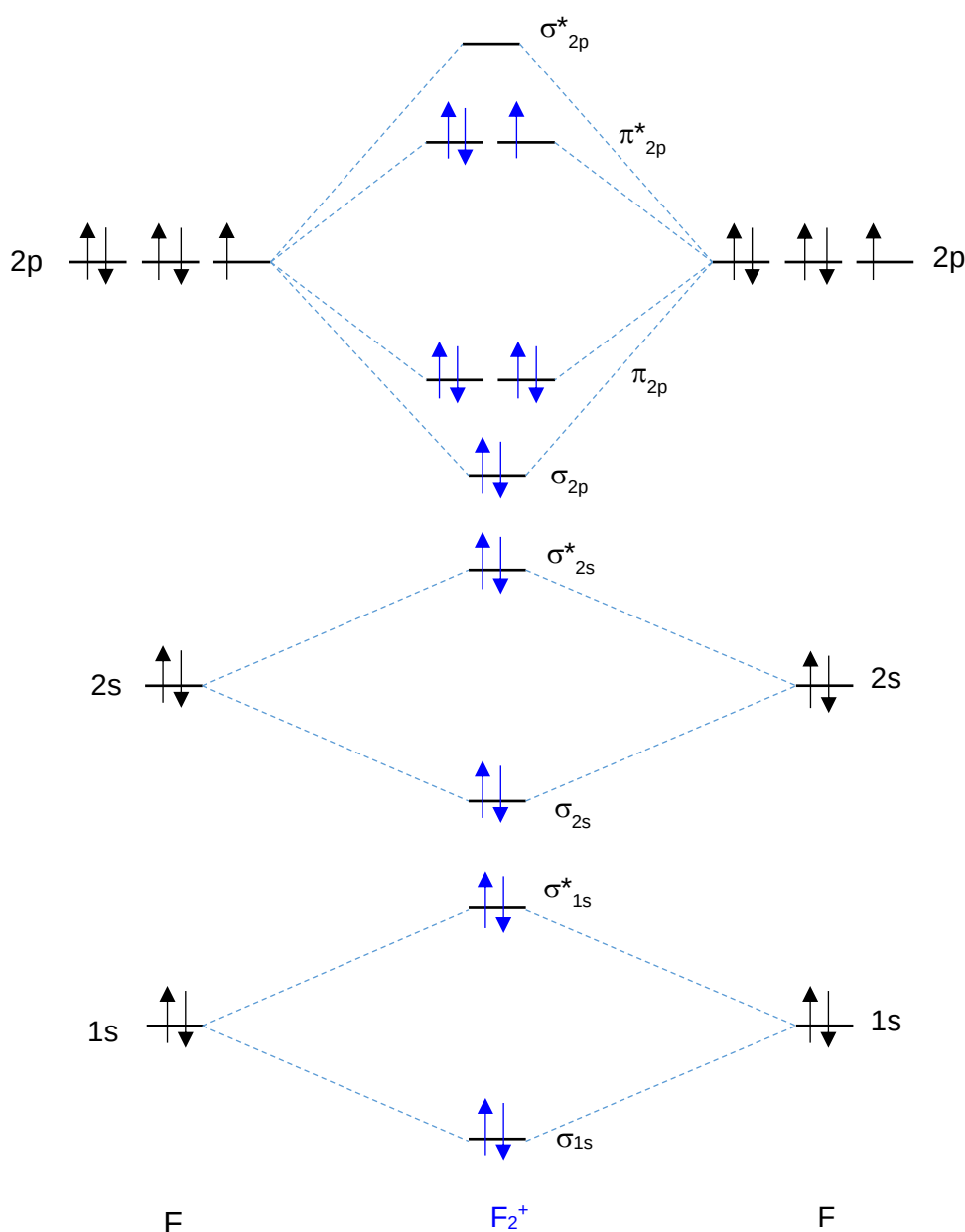
(g) (i) The reaction require light to induce the reactant/catalyst to form an excited state and transfer of electron takes place (or cause an increase and decrease in oxidation state in certain atoms).

- (ii) Red light has a longer wavelength (or lower frequency) and may not provide sufficient energy to cause an excitation of reactants/catalyst.
- (iii) When R_1 is an aromatic ring, the p orbital of the radical carbon atom in the α -amino radical can overlap with the π electron cloud of aromatic ring, delocalising the unpaired electron into the aromatic ring and stabilise the α -amino radical.

When R_2 is an aromatic ring, the p orbital of the carbanion in the intermediate anion can overlap with the π electron cloud of aromatic ring, delocalising the lone pair of electrons into the aromatic ring and disperse the negative charge. This stabilises the intermediate anion.

Q2 Setter: ASRJC

2 (a) (i) MO diagram for F_2^+

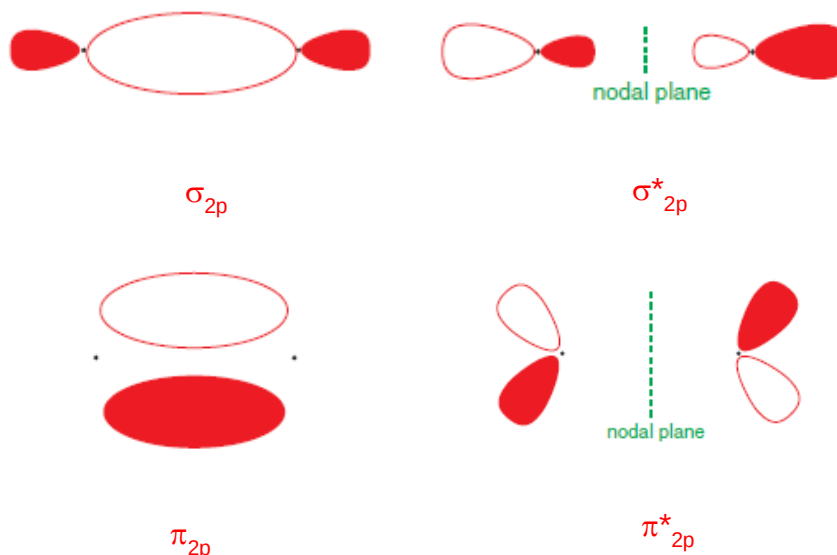


$$\text{Bond order of } F_2^+ = \frac{1}{2}(10 - 7) = \underline{1.5}$$

- (ii) LUMO is σ_{2p}^*
HOMO is π_{2p}^*

(iii) Paramagnetic as there is an unpaired electron (in the HOMO).

(b)



- (c) (i) When a photon of UV/VIS light is absorbed by an organic molecule which contains a chromophore, an electron gains the photon's energy and is promoted to a higher energy orbital.

Absorption in the UV/VIS region corresponds to $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$ and $n \rightarrow \sigma^*$ transitions.

(ii) Benzene/phenol and alkene.

(iii) The extended conjugation of the double bond with benzene ring would lead to longer wavelengths being absorbed for Δ^2 -THC.

(iv) Concentration of enantiomer in diluted drip solution
 $= A/\epsilon l = 0.954 / (2010 \times 1)$
 $= 4.75 \times 10^{-4} \text{ mol dm}^{-3}$

Concentration of enantiomer in drip solution
 $= 4.75 \times 10^{-4} \times 5 \times 5$
 $= 1.1875 \times 10^{-2} \text{ mol dm}^{-3}$
 $= \underline{1.19 \times 10^{-2} \text{ mol dm}^{-3}} \text{ (3 s.f.)}$

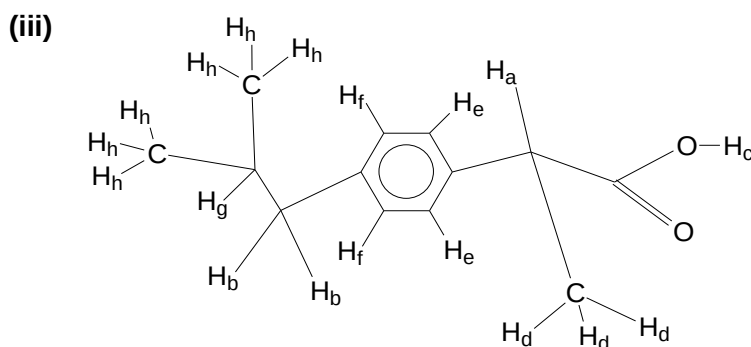
Q3 Setter: ASRJC

- 3 (a) (i) In an external applied magnetic field, all protons have their magnetic moments either parallel with the field or antiparallel to it. As a result, the degenerate spin states split into two states of unequal energy.

The energy difference between the two levels is small and falls within the radio frequency region.

The nuclear magnetic resonance phenomenon occurs when nuclei parallel with an applied magnetic field are induced to absorb energy and change their spin orientation such that it is antiparallel to the applied magnetic field.

- (ii) TMS is used as an internal reference as it is volatile and inert. The four carbon atoms attached to silicon in TMS are all equivalent. Silicon is more electropositive than carbon. Hence, the carbon are fairly electron-rich (or shielded), which means they resonate at frequency a little less than that of most organic compounds. Hence, it produces a strong singlet peak at a lower frequency than most ^1H absorptions in organic molecules, so its peak does not interfere with the other peaks. It will show a sharp singlet assigned $\delta = 0$.



- (iv) H_a will appear as a quartet since it is coupled to a CH_3 group. H_b will appear as a doublet since it is coupled to a CH group.

$\delta\text{H}_a = 3.5$ (proton is deshielded by $\text{C}=\text{O}$ as well as benzene ring)

$\delta\text{H}_b = 2.5$ (proton is deshielded by benzene ring) (accept 2.3 to 3.0)

Accept $\delta\text{H}_a > \delta\text{H}_b$

Accept δH_b between 2.3 and 3.0

(b)

$$x = \frac{100}{1.1} \left(\frac{2.96}{29.9} \right) = 9$$

Compound **E** consists of 9 carbons.

E reacts with aqueous bromine $\Rightarrow$ phenol or phenylamine must be present. Since ratio is 1:3 $\Rightarrow$ 2, 4 and 6 positions with respect to OH or NH_2 are not substituted by any group $\Rightarrow$ the two substituents in the benzene are in position 1 and 3 with respect to OH or NH_2 group.

From the IR spectrum:

Wavenumber	Deductions
Strong peak at 1710 to 1750 cm^{-1}	Presence of $\text{C}=\text{O}$ stretches of ester
Strong peak at 1200 to 1350 cm^{-1}	Presence of $\text{C}-\text{O}$ stretch of ester
Weak peak at 3300 to 3500 cm^{-1}	Presence of two $\text{N}-\text{H}$ stretches. Primary amine present.

From the NMR spectrum:

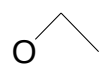
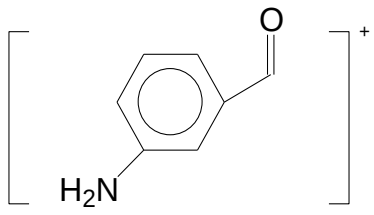
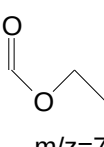
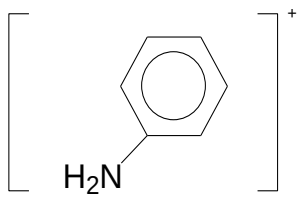
	δ	splitting pattern	number of protons	deduction
1	1.40	triplet	3	-CH ₂ -CH ₃
2	3.80	singlet	2	-NH ₂ (labile protons)
3	4.30	quartet	2	-CH ₂ -CH ₃ , next to electronegative O atom
4	6.84 – 7.42	multiplet	4	Benzene ring with 2 substituents

From the IR spectrum, it is confirmed that there is an ester group present, hence, from the M_r of **E**:

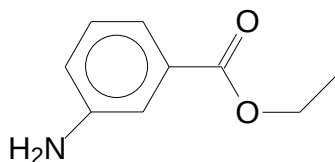
$165 - 12 \times 9 - 2 \times 16 = 25$ which corresponds to M_r of 1 nitrogen and 11 hydrogen. Hence, molecular formula of **E** is C₉H₁₁NO₂.

Or

$165 - (12 \times 9) - 11 = 46$. This corresponds to M_r of 1 nitrogen and 2 oxygen. Hence, molecular formula of **E** is C₉H₁₁NO₂.

m/z	Deduction	Fragment ions
120	$165 - 45 = 120$ $M^+ -$  m/z=45	
92	$165 - 73 = 92$ $M^+ -$  m/z=73	

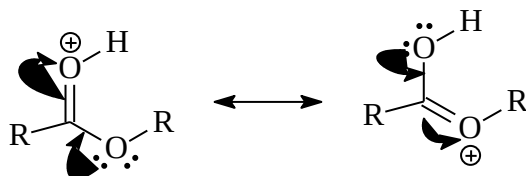
Structure of **E**:



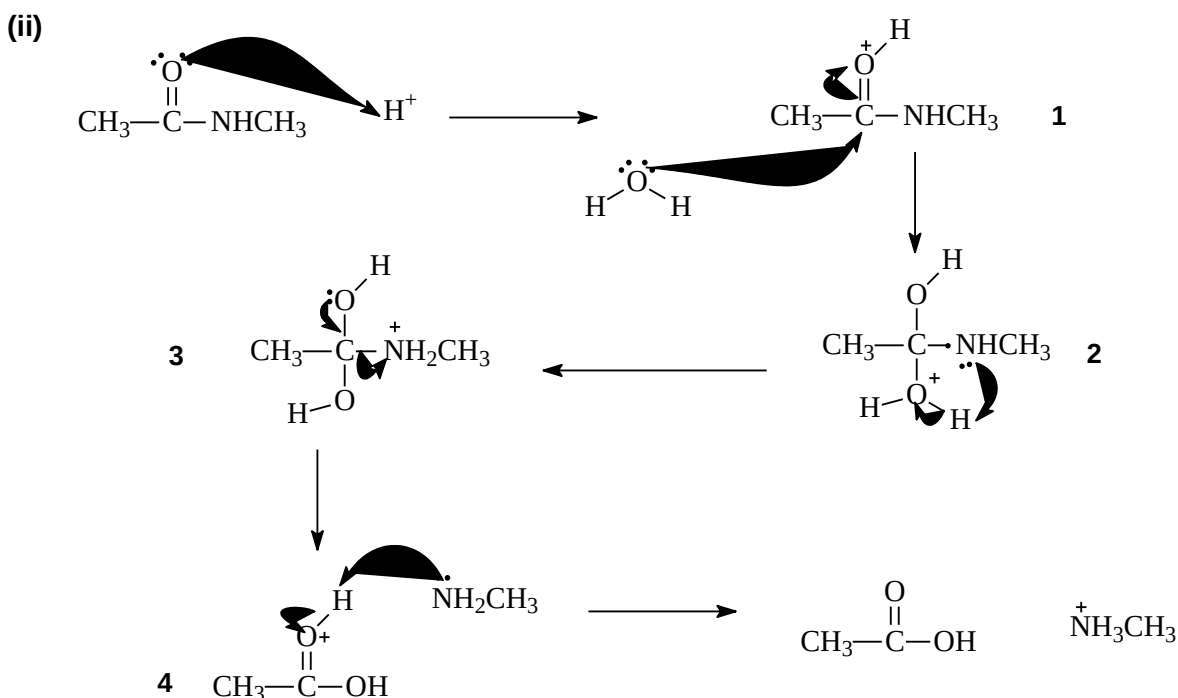
Q4 Setter: ASRJC

- 4 (a) (i) In cation **A**, the lone pair of electron on O is delocalised to the π electron cloud of the C=O. This disperses the positive charge on the O in the C=O bond and stabilises the cation.

Diagram is not required



cation **A**



- (iii) The lone pair of electron on O and N in ester and amide respectively can be delocalised to the π electron cloud of the C=O. Since N in amide is less electronegative than O in ester, the lone pair of electron on N in amide can be delocalised to a greater extent as compared to that of O in ester.

Hence, the carbonyl C in amide is less electron deficient and is less susceptible to a reaction with the nucleophile H_2O .

- (b) (i) The $\text{C}(\text{CH}_3)_3$ in **Q** poses a greater steric hindrance for the nucleophile OH^- to approach the carbonyl C in **Q**. This results in a greater activation energy and hence a smaller k .
- (ii) The value of k for ester **S** will be larger than that of ester **R**.

Since NO_2 is an electron-withdrawing group, the lone pair of electron on O is delocalised to the π electron cloud of the benzene to a greater extent. This will make the carbonyl C in **S** more electron deficient as compared to that of **R**. Hence it will be more susceptible to a reaction with the nucleophile OH^- , resulting in a smaller activation energy and hence a larger k .

(c) (i) Using the equation $E_a = A + B\Delta H$,

For ethylbenzoate: $+55.4 = A + B(+52.8)$ -----(1)

For ethyl 2-nitrobenzoate: $+48.6 = A + B(+46.0)$ -----(2)

Taking (1) – (2): $(+55.4 - (+48.6)) = B(+52.8 - (+46.0))$

$$6.8 = 6.8B$$

$$B = \underline{1.00}$$

Substituting $B = 1.00$ into equation (1)

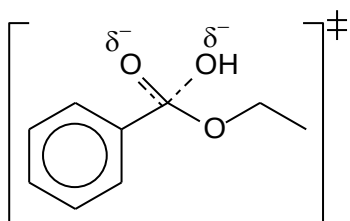
$$+55.4 = A + 1.00(52.8)$$

$$A = \underline{2.60}$$

For ethyl 4-nitrobenzoate: $E_a = 2.60 + 1.00(+45.8)$

$$E_a = \underline{+48.4 \text{ kJ mol}^{-1}}$$

(ii)



Section B

Q5 Setter: ASRJC

5 (a) (i) To 1 cm³ of the reaction mixture in a test tube, add 2,4-DNPH and warm.
If **A** is still present, orange ppt is formed.
If only **B** is present, no ppt is formed.

(ii) **B**: Peak at 1050 – 1330 cm⁻¹ due to the presence of ester C–O bond.

A: No peak at 1050 – 1330 cm⁻¹.

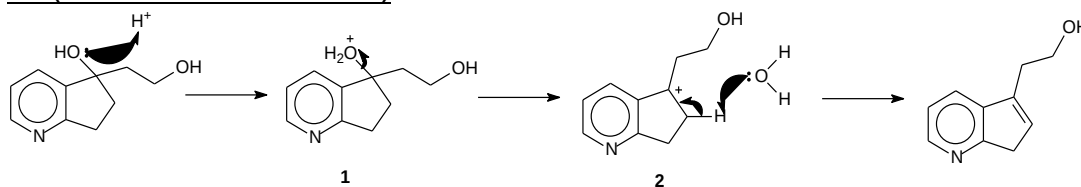
Or

B: Peak at 3200 – 3600 cm⁻¹ due to the presence of alcohol O–H bond.

A: No peak at 3200 – 3600 cm⁻¹.

(iii) LiAlH₄

(iv) E1 (Unimolecular elimination)



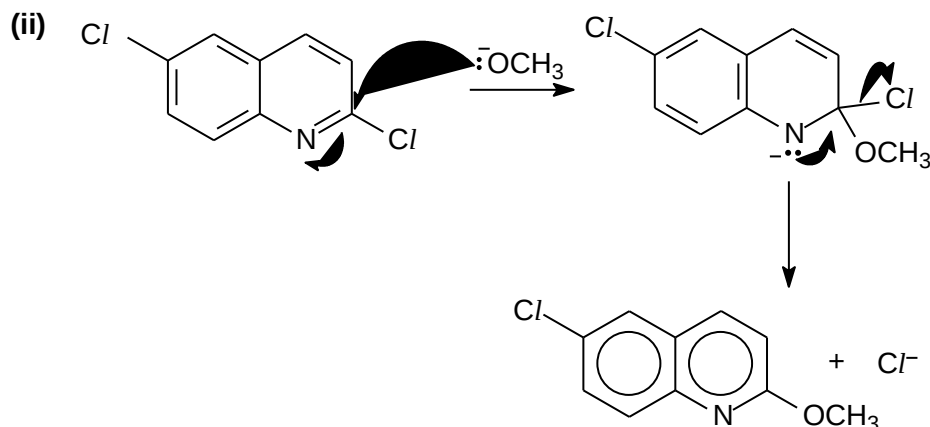
(b) (i) N1, N2: sp²

(ii) The lone pair of electrons on N1 is in a sp² hybrid orbital whereas the lone pair of electron in a secondary amine is in a sp³ hybrid orbital. The greater s character of a sp² orbital will cause the lone pair of electrons on N1 to be more strongly attracted to the nucleus, hence less available for donation.

(iii) The lone pair of electron in N2 is delocalised to the π electron of the pyrrole ring while the lone pair of electron in N1 is in a sp² hybrid orbital that is not delocalised to the π electron cloud of the pyridine ring. Hence the lone pair on N2 is less available for donation.

(c) (i) N is highly electronegative which exerts an electron withdrawing effect on the pyridine ring. This decreases the electron density in the pyridine ring and makes pyridine less susceptible to a reaction with the electrophile Br⁺ as compared to benzene.

The lone pair of electrons on N is not delocalised and acts a nucleophile to attack Br⁺. Hence the pyridine ring is not reactive.



(d) (i)

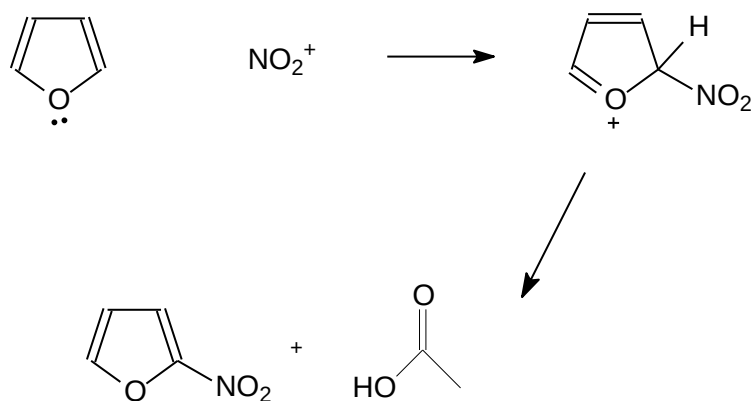
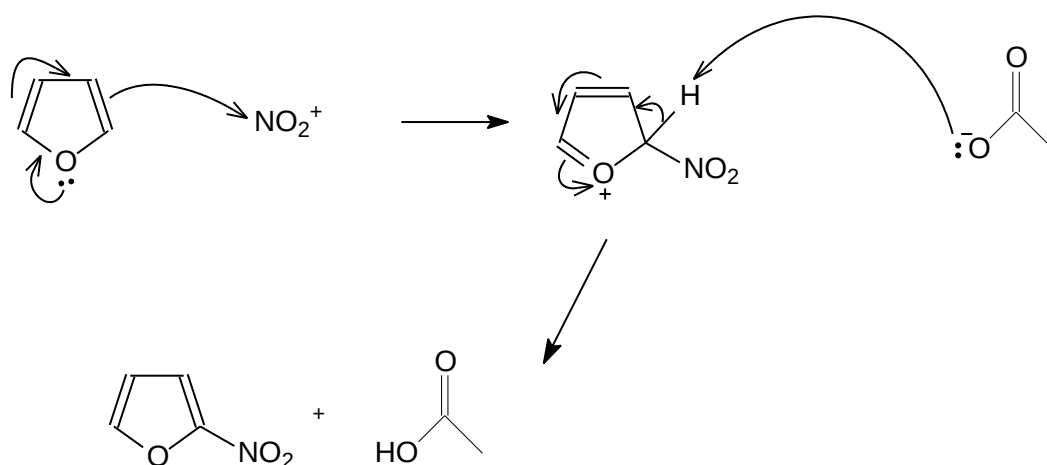
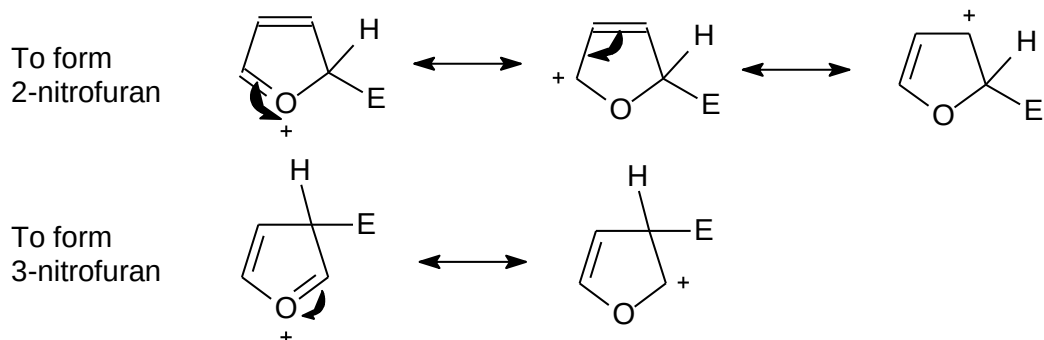


Fig 5.2



(ii)



Since there are more resonance structures for the intermediate of 2-nitrofuran, the intermediate is more stable and hence 2-nitrofuran is the major product.

(iii) Pyrrole has a faster rate of nitration than furan.

The N in pyrrole is less electronegative as compared to the O in furan. Hence it can delocalise its lone pair of electron to greater extent as compared to that of O in furan. This stabilises the cationic intermediate to a greater extent.

Or

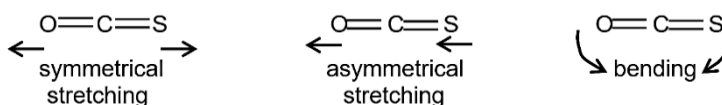
The N in pyrrole is less electronegative as compared to the O in furan. Hence the ring is more electron rich and more susceptible to a reaction with the electrophile.

Q6 Setter: NYJC

- 6 (a) (i) Molecules are able to absorb energy from the photons in the infrared region. The absorption of IR radiation is associated with the vibrational energy levels within a molecule. The different vibrational energy levels corresponds to the different vibrational modes of the molecule, such as stretching and bending of their covalent bonds.

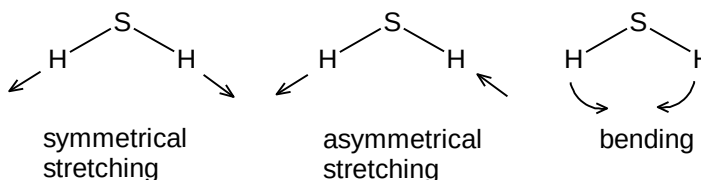
Only covalent bonds that have a dipole moment and have vibrations that result in a change of dipole moments are infrared active.

- (ii) OCS has a linear structure and the following modes of vibration.



Since O and S have different electronegativities, all the three vibrational modes result in a change in dipole moment and are infrared active. Hence, three peaks will be shown in the IR spectrum of OCS.

For a non-linear molecule such as H_2S , the vibrational modes are:



All the three vibrational modes are infrared active and three peaks will be shown in the IR spectrum.

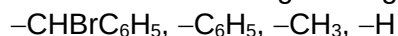
- (iii) Absorbance frequency is proportional to the bond strength. Since the bond energy of C=S (573 kJ mol^{-1}) is larger than that of S-H bond (347 kJ mol^{-1}), C=S bond will be stronger than S-H bond and its IR absorption frequency will be greater.

- (b) (i) The substituents about carbon 1 has the following ranking from highest to lowest:
 $-\text{Br}$, $-\text{C}_6\text{H}_5$, $-\text{CH}_2\text{C}_6\text{H}_5$, $-\text{H}$

The first, second and third ranking groups follow the order in a clockwise direction with H atom being directed into the plane of the paper.

Hence, carbon atom 1 in **F** has the *R* configuration.

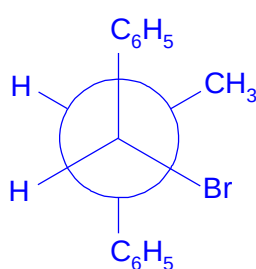
The substituents about carbon 2 has the following ranking from highest to lowest:



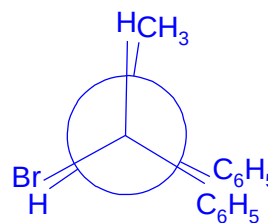
The first, second and third ranking groups follow the order in an anticlockwise direction with H atom being directed into the plane of the paper.

Hence, carbon atom 2 in **F** has the *S* configuration.

(ii)

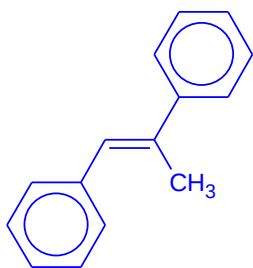
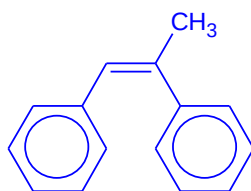


Gauche conformation
to produce alkene **G**



Eclipsed conformation
to produce alkene **H**

(iii)

Alkene **G**Alkene **H**

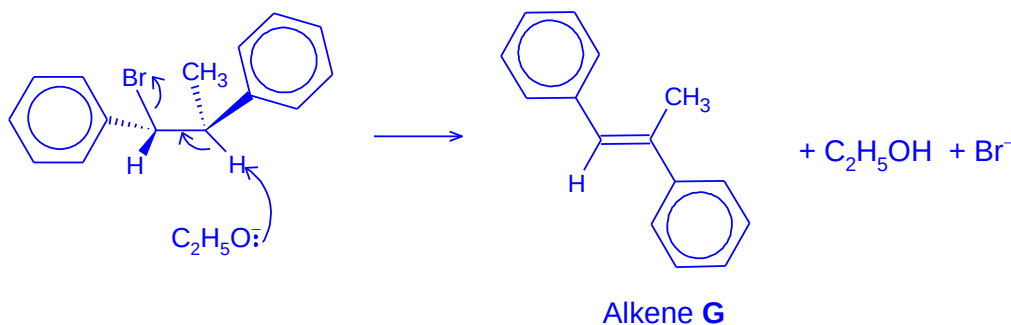
Alkene **G**: *E*; Alkene **H**: *Z*

- (iv) In the eclipsed conformation to produce alkene **H**, there is a significant steric strain and torsional strain between the $C_6H_5 \leftrightarrow C_6H_5$ interaction. Hence, this conformation is of higher energy and less stable.

In the gauche conformation to produce alkene **G**, the gauche interaction between the $C_6H_5 \leftrightarrow Br$ and $C_6H_5 \leftrightarrow CH_3$ interactions are less significant. Hence, this conformation is of lower energy and more stable.

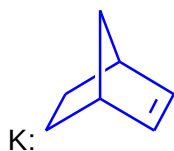
Therefore, alkene **G** will have a higher yield than alkene **H**.

(v)



- (c) (i) The bridgehead carbon cannot form the required planar arrangement for the carbocation, hence E1 elimination cannot occur.

(ii)



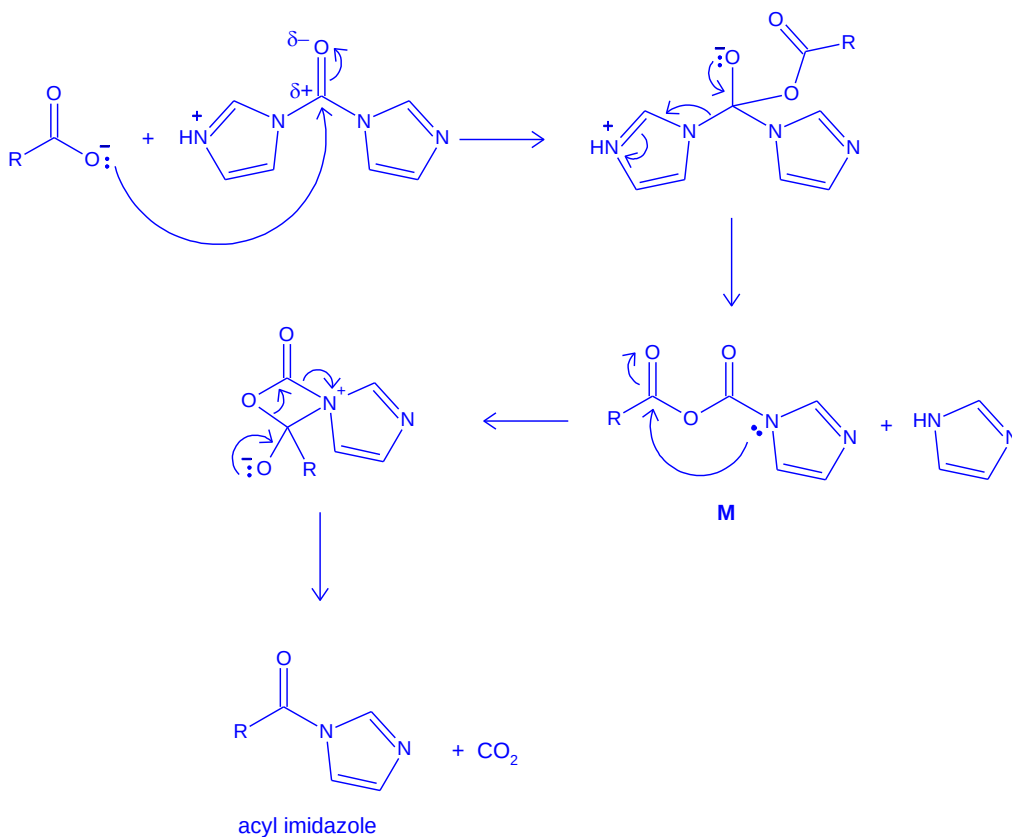
The elimination occurs from a syn periplanar arrangement of the C–D and C–Br bonds. E2 elimination from an anti periplanar arrangement cannot occur as the C–H bond available for elimination is at a 120° dihedral angle to the C–Br bond, which is energetically unfavourable.

- (d) Aqueous NaOH can act as both a strong nucleophile and a strong base. Hence, it is able to react with chlorocyclohexane via S_N2 and E2 to produce cyclohexanol and cyclohexene respectively.

Compound L can act as a strong base but weak nucleophile and react with chlorocyclohexane via E2 to produce cyclohexene. However, it is sterically hindered by bulky benzene rings. Hence, it is unable to act as a nucleophile to react with the electron deficient carbon via S_N2 .

Q7 Setter: NYJC

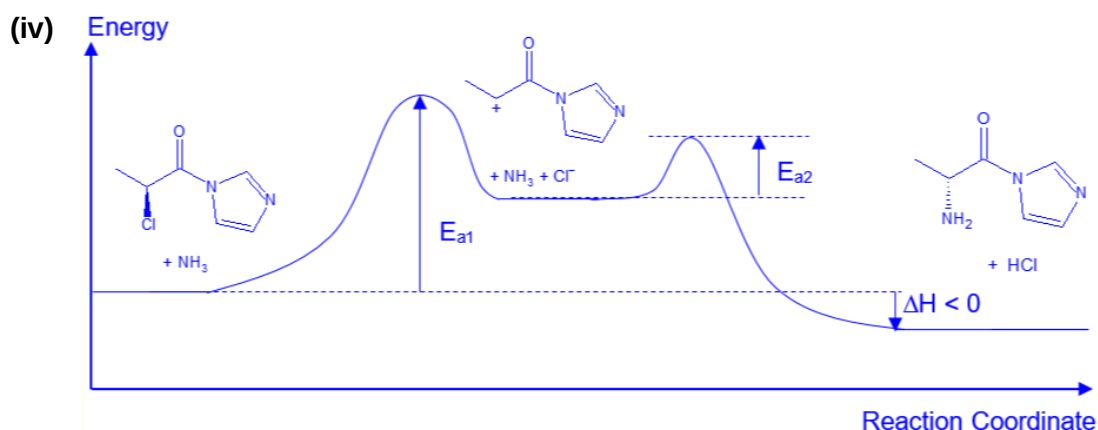
7 (a) (i)



- (ii) Acyl chlorides are sensitive to aqueous condition so it is able to react with water to give the carboxylic acids. However, acyl imidazoles are stable in the presence of water.
- (b) (i) Rate of reaction remains constant as the reaction undergoes S_N1 mechanism.
(i.e. Rate = $k[\text{N}]$)

(ii) The Cl^- leaving group is attracted to the carbocation even after cleavage, which prevents complete solvation and separation of the ions. This intimate ion pair recombines to give the original reactant; this is also known as internal return. As a result, this causes steric hindrance on one face by the Cl^- leaving group, favouring back-side attack, thus giving a higher percentage yield of one of the enantiomers.

(iii) Enantiomeric excess = $59 - 41 = 18\%$
 Specific rotation of minor product = $\frac{8.9}{18} \times 100 = +49.4^\circ$



(c) (i) Diastereomers

(ii) The halonium ion formed as an intermediate causes steric hindrance on one side of the molecule. Reaction with X^- ion in the second step could then occur only from the opposite side with no steric hindrance.

(iii) Hydrogen is a period 1 element and does not have energetically low-lying vacant orbitals to accommodate more electrons to form the three-membered ring structure like halonium ion.

Hence, electrophilic addition of hydrogen halide takes place via the carbocation intermediate, where the halide ion can attack the trigonal planar carbocation intermediate from the top and bottom of the plane at equal probability. This results in a racemic mixture being formed.

(iv) $[M] : [M+2]$ peaks of **Y** have a relative abundance of approximately 3:1. Hence **W** is chlorine.

$[M] : [M+2] : [M+4]$ peaks of **Z** have a relative abundance of approximately 3:4:1. Hence, **Z** contains one chlorine and one bromine. **V** is bromine.

(v) Other groups added into **Y** = $148 - 7(12.0) - 12(1.0) - 35.0 = 17.0$ (presence of OH group)

Hence, molecular formula of **Y** = $\text{C}_7\text{H}_{13}\text{OCl}$

Aqueous chlorine, room temperature

